

Deviation Is Data

A Cross-Sector Normalization Audit of Physical Laws

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Abstract

We propose a deliberately simple diagnostic for physical equations: apply one frozen, lossless time-to-length normalization, reduce each law only by declared invertible transformations, and treat whatever survives as the result rather than as an inconvenience to be removed. The base map is $w = ct$. It is elementary and is not claimed as new. The proposed contribution is the *cross-sector audit*: the same transformation grammar is applied to deliberately heterogeneous laws, including controls that should not collapse. Across mechanics, electromagnetism, circuits, quantum theory, transport, continuum physics, gravitation, cosmology, and static relations, the procedure separates coefficient conventions from persistent structure such as derivative order, characteristic scales, dimensionless couplings, nonlinearity, dissipation, dispersion, discreteness, sources, metric signature, and topology. Static laws are unchanged and therefore serve as negative controls. The method makes no ontological claim and derives no new constant. Its empirical content is prospective: if a small frozen grammar yields a stable residual taxonomy and organizes blinded holdout equations without retuning, that constitutes nontrivial representational compression; if the residual vocabulary proliferates or discretionary transformations are repeatedly required, the program fails to compress.

1. Question

Dimensional analysis, natural units, similarity theory, and symmetry reduction are standard tools. Writing time as ct is standard in relativity. The question here is therefore not whether the substitution

$$w = ct \tag{1}$$

is new. It is whether a *single frozen normalization protocol*, applied without sector-by-sector cosmetic retuning, provides a useful classification of physical laws by the structure that cannot be normalized away.

The operative principle is:

Normalize without loss; record the residue. Deviation is data.

The desired output is not “one equation for all physics.” It is a map of exactly how far different systems are equivalent under a declared transformation class, and exactly what prevents further equivalence.

2. Protocol

Let E denote a physical equation and $\mathcal{T}$ the declared class of admissible transformations.

Base map. Apply $\mathcal{N}_c : t \mapsto w = ct$ wherever a time coordinate appears. Then

$$\partial_t = c \partial_w, \quad \partial_t^n = c^n \partial_w^n, \quad \frac{dx}{dw} = \frac{v}{c}. \tag{2}$$

The value of c is frozen after calibration and is not re-fit equation by equation.

Secondary maps. Further transformations are allowed only if they are (i) invertible, (ii) explicitly recorded, and (iii) universal or algebraically determined by parameters already present in the law. Characteristic scales may be extracted and then scaled out, but the extracted scale remains part of the ledger.

Residual. Define the audit residual only relative to the declared transformation class,

$$\mathcal{R}(E \mid \mathcal{T}) = \text{information in } E \text{ that remains after canonicalization under } \mathcal{T}. \tag{3}$$

Typical residuals include derivative-order mismatch, characteristic length or inverse length, dimensionless coupling, signature, nonlinearity, dissipation, dispersion, finite spacing, source structure, boundary/topological data, or the absence of a temporal variable.

Failure condition. If each sector requires a discretionary new map, the exercise reduces to ordinary problem-specific nondimensionalization. The method earns value only if the transformation grammar remains small while the residual classes remain stable across heterogeneous laws and, ultimately, blinded holdouts.

3. Calibration and representative outcomes

3.1 Positive calibrators

For a trajectory $v = dx/dt$, equation (1) gives

$$\beta = \frac{dx}{dw} = \frac{v}{c}, \quad (4)$$

so a c -propagating trajectory has unit coordinate slope. For

$$\partial_t^2 \psi = u^2 \nabla^2 \psi, \quad (5)$$

we obtain

$$\partial_w^2 \psi = \left(\frac{u}{c}\right)^2 \nabla^2 \psi. \quad (6)$$

For $u = c$, the derivative coefficients balance. The Lorentzian sign remains when the operator is written as $\partial_w^2 - \nabla^2$; coefficient balance does not erase metric signature.

Vacuum Maxwell theory provides a first-order positive control. With $\tilde{\mathbf{B}} = c\mathbf{B}$,

$$\nabla \cdot \mathbf{E} = 0, \quad \nabla \cdot \tilde{\mathbf{B}} = 0, \quad \partial_w \mathbf{E} = \nabla \times \tilde{\mathbf{B}}, \quad \partial_w \tilde{\mathbf{B}} = -\nabla \times \mathbf{E}. \quad (7)$$

This is a standard coefficient-balanced reformulation, not a new field theory.

3.2 Structured residues

The free Schrödinger equation becomes

$$i\partial_w \psi = -\frac{\bar{\lambda}_C}{2} \nabla^2 \psi, \quad \bar{\lambda}_C = \frac{\hbar}{mc}, \quad (8)$$

so the mass parameter appears as a length while first-order temporal evolution remains first order. Hydrogen retains the dimensionless fine-structure constant α after the Compton scale is removed; the normalization does not determine its value.

Diffusion becomes

$$\partial_w T = \frac{D}{c} \nabla^2 T, \quad (9)$$

retaining both the first-versus-second derivative-order distinction and the transport length D/c . A nonlinear pendulum, after its recurrence scale is removed, retains $\sin \theta - \theta$ as a finite-amplitude residue. A discrete oscillator chain retains its finite-spacing sine dispersion. Newtonian gravity and Coulomb attraction reduce to the same inverse-square dimensionless central-force form after their distinct interaction lengths are scaled out, without implying that the forces are physically identical.

4. Controls: where the procedure should not “work”

The audit is designed to fail cleanly.

Partial controls. Radioactive decay, diffusion, Fokker–Planck, Navier–Stokes, Boltzmann, Korteweg–de Vries, Allen–Cahn, and Cahn–Hilliard equations all retain characteristic operator structure. The substitution $w = ct$ changes the dimensional bookkeeping but does not change first order into second order, remove collisions, erase nonlinear advection, or eliminate dispersion.

Negative controls. Static equations with no time variable are unchanged by the base map: Laplace and Poisson equations, the ideal-gas law, the first law of thermodynamics, Boltzmann entropy, static Hooke law, and algebraic equations of state. The method therefore does not manufacture temporal geometry where none occurs in the law.

These failures are not exceptions to the method; they are its primary observations.

5. What the audit measures

Across the battery, familiar dimensional conversions recur:

$$\omega \mapsto \frac{\omega}{c}, \quad \tau \mapsto c\tau, \quad D \mapsto \frac{D}{c}, \quad \frac{\hbar}{m} \mapsto \frac{\hbar}{mc}, \quad GM \mapsto \frac{GM}{c^2}. \quad (10)$$

These identities are not the proposed result. The result is the residual pattern after such conventional freedom has been factored out.

The working conjecture is intentionally modest:

A common lossless normalization may provide a useful cross-sector coordinate system for classifying physical laws by the mathematical structures that survive it.

The conjecture becomes interesting only if the residual vocabulary remains unexpectedly small, stable, and predictive on equations not used to formulate the taxonomy.

6. Prior art and scope

The components of the procedure have substantial precedent. Buckingham formalized dimensionless groups and similitude [1]; Bridgman systematized dimensional analysis [2]; Wolsky examined uniform changes of length and time scales across classical mechanics, nonrelativistic quantum mechanics, and relativistic quantum field theory [3]. Natural-unit and geometrized-unit methods routinely identify time and length through c ; Gilliard explicitly rewrote broad classes of equations so that c and t enter through ct [6]. Lie-symmetry methods formalize nondimensionalization through invariant coordinates [5], and modern data-driven work searches for dimensionless groups and governing laws under Buckingham-II constraints [7, 8].

Mukohyama and Uzan address a distinct but nearby problem: they begin with a positive-definite four-dimensional metric and introduce a scalar clock field whose gradient selects a local time direction; tuned couplings then yield effective Lorentzian dynamics for classical fields [4]. The present audit introduces no clock field and modifies no dynamics. It starts from established equations and asks only what a frozen lossless normalization removes or leaves behind.

Accordingly, no novelty is claimed for ct coordinates, natural units, nondimensionalization, similarity theory, or Euclidean-to-Lorentzian constructions. The proposed contribution is the protocol: broad cross-sector use of a frozen base map, an explicit transformation ledger, deliberately non-collapsing controls, and success criteria based on compression and out-of-sample organization.

7. Falsifiable next step

The next phase should be computational and preregistered. A parser should accept a held-out battery of equations and output: (i) the normalized form; (ii) characteristic scales implied by the original coefficients; (iii) dimensionless groups; (iv) operator-order and algebraic structure; and (v) a residual class. The allowed transformation grammar should be frozen before scoring the holdout set.

Three outcomes are then distinguishable:

1. **No compression:** residual classes proliferate or require discretionary retuning.
2. **Representational compression:** a small fixed grammar and a small stable residual vocabulary organize many sectors.
3. **Stronger structure:** residual relations inferred from one family correctly organize quantitative relations in independent holdout families without additional fitting.

Only the third outcome would motivate claims beyond methodology.

8. Conclusion

The map $w = ct$ is elementary. Used once, it is mostly a familiar change of units. Used as a frozen cross-sector audit, it poses a disciplined question: *what disappears, what survives, and do the survivors recur in a small number of*

forms? The demonstrations below show both positive and negative behavior. That is the intended result. The method is not designed to make every equation fit; it is designed to turn the pattern of fit and non-fit into structured data.

A. Broad stress-test battery

Table 1: Representative cross-sector audit. The table is intentionally heterogeneous; static and non-collapsing laws are included as controls.

Law	Starting form	After $w = ct$ and declared secondary maps	Primary residue
Kinematics	$v = dx/dt$	$dx/dw = v/c \equiv \beta$	dimensionless slope β
Acceleration	$a = d^2x/dt^2$	$x_{ww} = a/c^2$	curvature-like inverse length
Newton II	$m\ddot{x} = F$	$x_{ww} = F/(mc^2)$	force relative to rest-energy scale
Euler-Lagrange	$\frac{d}{dt}L\dot{q} - L_q = 0$	with $\tilde{L}(q, q_w, w) = L(q, cq_w, w/c)$, the variational form is preserved	transformed Lagrangian; no universal collapse
Hamilton equations	$\dot{q} = H_p, \dot{p} = -H_q$	$q_w = H_p/c, p_w = -H_q/c$	Hamiltonian scale remains
Lorentz force	$m\dot{\mathbf{v}} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$	$mc^2\mathbf{U}_w = q(\mathbf{E} + \mathbf{U} \times \tilde{\mathbf{B}}), \mathbf{U} = \mathbf{v}/c, \tilde{\mathbf{B}} = c\mathbf{B}$	field coupling inverse lengths
Simple harmonic oscillator	$\ddot{x} + \omega^2x = 0$	$x_{ww} + (\omega/c)^2x = 0$	$\ell_\omega = c/\omega$
Damped oscillator	$\ddot{x} + 2\gamma\dot{x} + \omega_0^2x = 0$	$x_{ww} + 2(\gamma/c)x_w + (\omega_0/c)^2x = 0$	two inverse lengths; γ/ω_0
Driven oscillator	$\ddot{x} + 2\gamma\dot{x} + \omega_0^2x = (F_0/m) \cos \Omega t$	RHS = $(F_0/mc^2) \cos[(\Omega/c)w]$	forcing curvature; frequency ratios
Pendulum	$\ddot{\theta} + (g/L) \sin \theta = 0$	$\theta_{ww} + [g/(Lc^2)] \sin \theta = 0$	$c\sqrt{L/g}$; nonlinear $\sin \theta$
Kepler orbit	$\ddot{\mathbf{r}} = -(GM/r^3)\mathbf{r}$	$\mathbf{r}_{ww} = -(GM/c^2)\mathbf{r}/r^3$	$\ell_G = GM/c^2$
Coulomb orbit	$\ddot{\mathbf{r}} = -[qQ/(4\pi\epsilon_0 m)]\mathbf{r}/r^3$	$\mathbf{r}_{ww} = -\ell_E\mathbf{r}/r^3$	$\ell_E = qQ /(4\pi\epsilon_0 mc^2)$
Discrete chain	$m\ddot{x}_n = K(x_{n+1} - 2x_n + x_{n-1})$	$x_{n,ww} = [K/(mc^2)]\Delta_d x_n$	spacing a ; sine dispersion
Euler-Bernoulli beam	$\rho A y_{tt} + EI y_{xxxx} = 0$	$y_{ww} + [EI/(\rho A c^2)] y_{xxxx} = 0$	fourth-order spatial operator; length ² coefficient
Linear elasticity	$\rho \mathbf{u}_{tt} = (\lambda + \mu)\nabla(\nabla \cdot \mathbf{u}) + \mu \nabla^2 \mathbf{u}$	divide by ρc^2	wave-speed ratios $c_L/c, c_T/c$
Generic wave	$\psi_{tt} = u^2 \nabla^2 \psi$	$\psi_{ww} = (u/c)^2 \nabla^2 \psi$	characteristic speed u/c
Vacuum Maxwell	$\nabla \times \mathbf{E} = -\mathbf{B}_t, \nabla \times \mathbf{B} = \mathbf{E}_t/c^2$	with $\tilde{\mathbf{B}} = c\mathbf{B}$: $\tilde{\mathbf{B}}_w = -\nabla \times \mathbf{E}, \mathbf{E}_w = \nabla \times \tilde{\mathbf{B}}$	Lorentzian field structure
Conducting EM wave	$\nabla^2 \mathbf{E} = \mu\epsilon \mathbf{E}_{tt} + \mu\sigma \mathbf{E}_t$	$\mathbf{E}_{ww} = [1/(\mu\epsilon c^2)]\nabla^2 \mathbf{E} - [\sigma/(\epsilon c)]\mathbf{E}_w$	speed ratio; attenuation inverse length
Telegrapher equation	$V_{xx} = LCV_{tt} + RCV_t$	$V_{ww} = [1/(LCc^2)]V_{xx} - [R/(Lc)]V_w$	propagation ratio; attenuation
RC circuit	$RC\dot{V} + V = 0$	$V_w + V/(cRC) = 0$	$\ell_{RC} = cRC$
LC circuit	$LC\ddot{Q} + Q = 0$	$Q_{ww} + Q/(c^2LC) = 0$	$\ell_{LC} = c\sqrt{LC}$
RLC circuit	$L\ddot{Q} + R\dot{Q} + Q/C = 0$	$Q_{ww} + [R/(Lc)]Q_w + [1/(LCc^2)]Q = 0$	damping and recurrence inverse lengths
Free Schrödinger	$i\hbar\psi_t = -(\hbar^2/2m)\nabla^2\psi$	$i\psi_w = -(\bar{\lambda}_C/2)\nabla^2\psi$	$\bar{\lambda}_C = \hbar/(mc)$; operator order

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Law	Starting form	After $w = ct$ and declared secondary maps	Primary residue
Time-dependent Schrödinger	$i\hbar\psi_t = [-(\hbar^2/2m)\nabla^2 + V]\psi$	$i\psi_w = [-(\bar{\lambda}_C/2)\nabla^2 + V/(\hbar c)]\psi$	Compton length; potential inverse length
Hydrogen	TDSE with $-e^2/(4\pi\epsilon_0 r)$	after $r = \bar{\lambda}_C \rho$: Coulomb term $-\alpha/\rho$	α ; Compton/Bohr scale ratio
Quantum oscillator	TDSE with $\frac{1}{2}m\omega^2 x^2$	natural $a_{\text{osc}}^2 = \bar{\lambda}_C(c/\omega) = \hbar/(m\omega)$	geometric-mean scale
Pauli equation	$i\hbar\psi_t = [(\mathbf{p} - q\mathbf{A})^2/(2m) + q\phi - (q\hbar/2m)\boldsymbol{\sigma} \cdot \mathbf{B}]\psi$	divide Hamiltonian by $\hbar c$ after $w = ct$	Compton scale; gauge and Zeeman couplings
Klein–Gordon	$c^{-2}\phi_{tt} - \nabla^2\phi + (mc/\hbar)^2\phi = 0$	$\phi_{ww} - \nabla^2\phi + \phi/\bar{\lambda}_C^2 = 0$	signature; inverse Compton mass scale
Dirac	$(i\hbar\gamma^\mu\partial_\mu - mc)\psi = 0$	with $x^0 = w$: $(i\gamma^\mu\partial_\mu - 1/\bar{\lambda}_C)\psi = 0$	Clifford/Lorentz structure; $1/\bar{\lambda}_C$
Proca	$\partial_\mu F^{\mu\nu} + (mc/\hbar)^2 A^\nu = 0$	$\partial_\mu F^{\mu\nu} + A^\nu/\bar{\lambda}_C^2 = 0$	mass length; covariant structure
Gross–Pitaevskii	$i\hbar\psi_t = [-(\hbar^2/2m)\nabla^2 + V + g \psi ^2]\psi$	$i\psi_w = [-(\bar{\lambda}_C/2)\nabla^2 + V/(\hbar c) + g \psi ^2/(\hbar c)]\psi$	nonlinear coupling
Planck–Einstein	$E = \hbar\omega$	$E/(\hbar c) = \omega/c$	algebraic inverse-length conversion; no base-time action
de Broglie	$p = \hbar k$	$p/\hbar = k$	algebraic inverse-length conversion
Relativistic dispersion	$E^2 - p^2 c^2 = m^2 c^4$	$\varepsilon^2 - \pi^2 = 1$, $\varepsilon = E/(mc^2)$, $\pi = p/(mc)$	Lorentzian sign
Advection	$\phi_t + \mathbf{u} \cdot \nabla \phi = 0$	with $\mathbf{U} = \mathbf{u}/c$: $\phi_w + \mathbf{U} \cdot \nabla \phi = 0$	dimensionless velocity field
Diffusion / heat	$T_t = D\nabla^2 T$	$T_w = (D/c)\nabla^2 T$	transport length; operator order
Reaction–diffusion	$u_t = D\nabla^2 u + R(u)$	$u_w = (D/c)\nabla^2 u + R(u)/c$	transport + reaction scales; nonlinearity
Radioactive decay	$\dot{N} = -\lambda N$	$N_w = -(\lambda/c)N$	decay inverse length; no spatial operator
Logistic growth	$\dot{N} = rN(1 - N/K)$	$N_w = (r/c)N(1 - N/K)$	growth inverse length; saturation nonlinearity
Fokker–Planck	$p_t = -\nabla \cdot (\mathbf{b}p) + D\nabla^2 p$	with $\mathbf{b} = c\mathbf{B}$: $p_w = -\nabla \cdot (\mathbf{B}p) + (D/c)\nabla^2 p$	drift field; diffusion length
Continuity	$\rho_t + \nabla \cdot (\rho\mathbf{u}) = 0$	$\rho_w + \nabla \cdot (\rho\mathbf{U}) = 0$	coefficient-balanced conservation law
Euler fluid	$\mathbf{u}_t + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p/\rho + \mathbf{f}$	$\mathbf{U}_w + (\mathbf{U} \cdot \nabla)\mathbf{U} = -\nabla[p/(\rho c^2)] + \mathbf{f}/c^2$	pressure ratio; forcing curvature
Navier–Stokes	$\mathbf{u}_t + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p/\rho + \nu\nabla^2\mathbf{u} + \mathbf{f}$	$\mathbf{U}_w + (\mathbf{U} \cdot \nabla)\mathbf{U} = -\nabla\Pi + (\nu/c)\nabla^2\mathbf{U} + \mathbf{f}/c^2$	viscosity length; nonlinear advection
Burgers	$u_t + uu_x = \nu u_{xx}$	$U_w + UU_x = (\nu/c)U_{xx}$	viscosity length; nonlinearity
Boltzmann	$f_t + \mathbf{v} \cdot \nabla_x f + (\mathbf{F}/m) \cdot \nabla_v f = C[f]$	with $\mathbf{v} = c\mathbf{U}$: $f_w + \mathbf{U} \cdot \nabla_x f + (\mathbf{a}/c^2) \cdot \nabla_U f = C[f]/c$	collision operator; phase-space structure
Korteweg–de Vries	$\eta_t + c_0\eta_x + \alpha\eta\eta_x + \beta\eta_{xxx} = 0$	$\eta_w + (c_0/c)\eta_x + (\alpha/c)\eta\eta_x + (\beta/c)\eta_{xxx} = 0$	mixed derivative order; nonlinear and dispersive scales
Sine–Gordon	$u_{tt} - v^2 u_{xx} + \omega_0^2 \sin u = 0$	$u_{ww} - (v/c)^2 u_{xx} + (\omega_0/c)^2 \sin u = 0$	speed ratio; nonlinear recurrence scale
Allen–Cahn	$u_t = D\nabla^2 u - f(u)$	$u_w = (D/c)\nabla^2 u - f(u)/c$	first-order time; nonlinear reaction
Cahn–Hilliard	$u_t = M\nabla^2[f'(u) - \kappa\nabla^2 u]$	$u_w = (M/c)\nabla^2[f'(u) - \kappa\nabla^2 u]$	second/fourth spatial order; nonlinear chemical potential

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Law	Starting form	After $w = ct$ and declared secondary maps	Primary residue
Porous-medium	$u_t = D\nabla^2(u^m)$	$u_w = (D/c)\nabla^2(u^m)$	nonlinear diffusion exponent m
MHD induction	$\mathbf{B}_t = \nabla \times (\mathbf{u} \times \mathbf{B}) + \eta \nabla^2 \mathbf{B}$	$\mathbf{B}_w = \nabla \times (\mathbf{U} \times \mathbf{B}) + (\eta/c)\nabla^2 \mathbf{B}$	magnetic diffusivity length
Newtonian Poisson	$\nabla^2 \Phi = 4\pi G\rho$	base map unchanged; optionally $\varphi = \Phi/c^2 \Rightarrow \nabla^2 \varphi = 4\pi G\rho/c^2$	static source structure
Electrostatic Poisson	$\nabla^2 V = -\rho_q/\epsilon_0$	unchanged	static negative control
Gauss electrostatics	$\nabla \cdot \mathbf{E} = \rho_q/\epsilon_0$	unchanged	static negative control
Laplace	$\nabla^2 \phi = 0$	unchanged	static negative control
Helmholtz	$\nabla^2 \psi + k^2 \psi = 0$	unchanged; if $k = \omega/u$, compare with ω/c	frequency-domain inverse length
Geodesic	$\ddot{x}^\mu + \Gamma_{\alpha\beta}^\mu \dot{x}^\alpha \dot{x}^\beta = 0$	with $s = c\tau$: $x^{\mu''} + \Gamma_{\alpha\beta}^\mu x^{\alpha'} x^{\beta'} = 0$	connection/metric already geometric
Einstein equation	$G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$	already geometrized under standard conventions	metric signature, curvature, Λ , matter tensor
Friedmann	$H^2 = 8\pi G\rho/3 - kc^2/a^2 + \Lambda c^2/3$	$(H/c)^2 = 8\pi G\rho/(3c^2) - k/a^2 + \Lambda/3$	density, curvature, Λ , all as inverse length ²
Curved-space scalar	$(\square - m^2 c^2/\hbar^2)\phi = 0$	mass term $1/\bar{\lambda}_C^2$; $x^0 = ct$ if chosen	metric and curvature dependence
Ideal gas	$PV = Nk_B T$	unchanged	no-time negative control
First law	$dU = TdS - pdV + \mu dN$	unchanged	no-time thermodynamic constraint
Boltzmann entropy	$S = k_B \ln \Omega$	unchanged	no-time statistical relation
Static Hooke law	$F = -kx$	unchanged	constitutive law; time enters only after dynamics
Equation of state	$p = w_{\text{eos}} \rho c^2$	$p/(\rho c^2) = w_{\text{eos}}$	already dimensionless constitutive parameter

B. Interpretive ledger

For clarity, the audit classifies outcomes rather than ranking them:

- *Balanced*: the fixed map produces equal or dimensionless coefficients without further discretionary choices (e.g. a c -wave, vacuum Maxwell with $\tilde{B} = cB$).
- *Scaled*: the map exposes a characteristic length, inverse length, or dimensionless ratio (e.g. D/c , ν/c , $\hbar/(mc)$, GM/c^2).
- *Structured residue*: derivative order, nonlinearity, dispersion, collision structure, signature, or topology persists.
- *Null action*: the base temporal map does nothing because the law contains no time variable.

A useful theory of the audit would explain why these classes recur and whether the list closes under a preregistered expansion of the battery.

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